

The background of the slide features a dark blue field with vertical columns of glowing binary digits (0s and 1s) in light blue and orange. Interspersed among the digits are soft, out-of-focus bokeh lights in shades of blue and orange, creating a digital, high-tech atmosphere.

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Let us look at a time-series dataset and examine more data exploration and cleaning steps



Time Series Example: AirPassenger Dataset

- The AirPassengers dataset contains the monthly totals of international airline passengers from January 1949 to December 1960.
- Rows: 144 (one per month)
- Columns:
 - **Month:** Date (formatted as YYYY-MM)
 - **Passengers:** Number of airline passengers that month (in thousands)



Data Exploration

```
[4] import pandas as pd
```

```
[5] from google.colab import files
    uploaded = files.upload()
```

Choose Files airline-passengers.csv

- airline-passengers.csv(text/csv) - 2180 bytes, last modified: 6/17/2025 - 100% done

Saving airline-passengers.csv to airline-passengers.csv

```
df = pd.read_csv('airline-passengers.csv')
# Preview
# Display the first 5 rows using iloc
print(df.iloc[0:5])
```

	Month	Passengers
0	1949-01	112
1	1949-02	118
2	1949-03	132
3	1949-04	129
4	1949-05	121

```
[8] # Display the 7th row (index position 4)
    print(df.iloc[6])
```

Month	1949-07
Passengers	148

Name: 6, dtype: object



Data Exploration

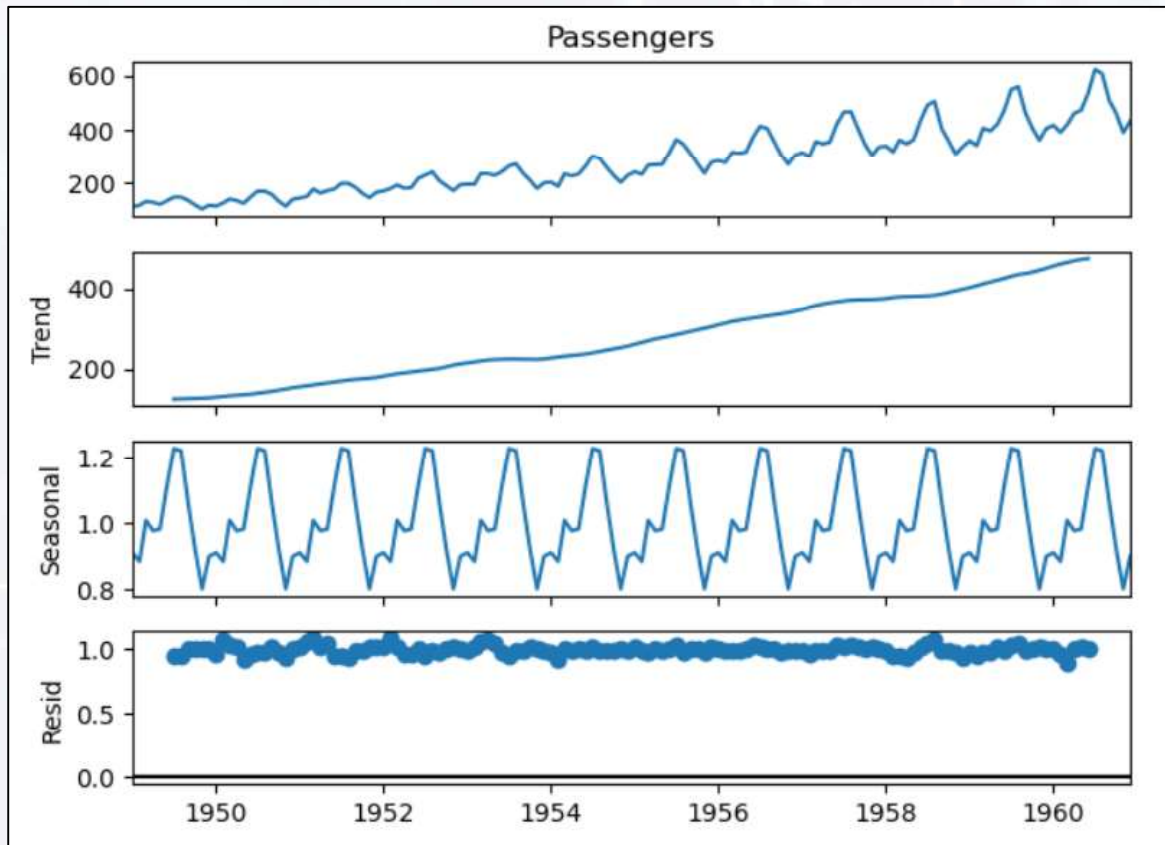
- Components:
 - Trend: Long-term direction in the data (e.g., steady increase)
 - Seasonality: Repeating patterns at regular intervals (e.g., yearly cycles)
 - Residual/Noise: Irregular, random fluctuations not explained by trend or seasonality
- This is part of **data exploration** — it helps you understand the structure and behavior of the time series before modeling or forecasting.
- Visualising the data helps to identify missing values or narrow down an infilling method.

Exploring Trends, Seasonality, and Noise

```
3]: from statsmodels.tsa.seasonal import seasonal_decompose  
  
result = seasonal_decompose(df['Passengers'], model='multiplicative')  
result.plot()  
plt.tight_layout()  
plt.show()
```




Data Exploration



- **Plot 1: Observed**

- The original data: monthly passenger counts from 1949 to 1960
- Shows an upward trend and seasonal fluctuations

- **Plot 2: Trend**

- The long-term direction in the data
- Passenger numbers are steadily increasing over time

- **Plot 3: Seasonal**

- Regular repeating patterns (seasonality)
- Clear yearly cycle: passenger numbers peak and drop at the same months each year

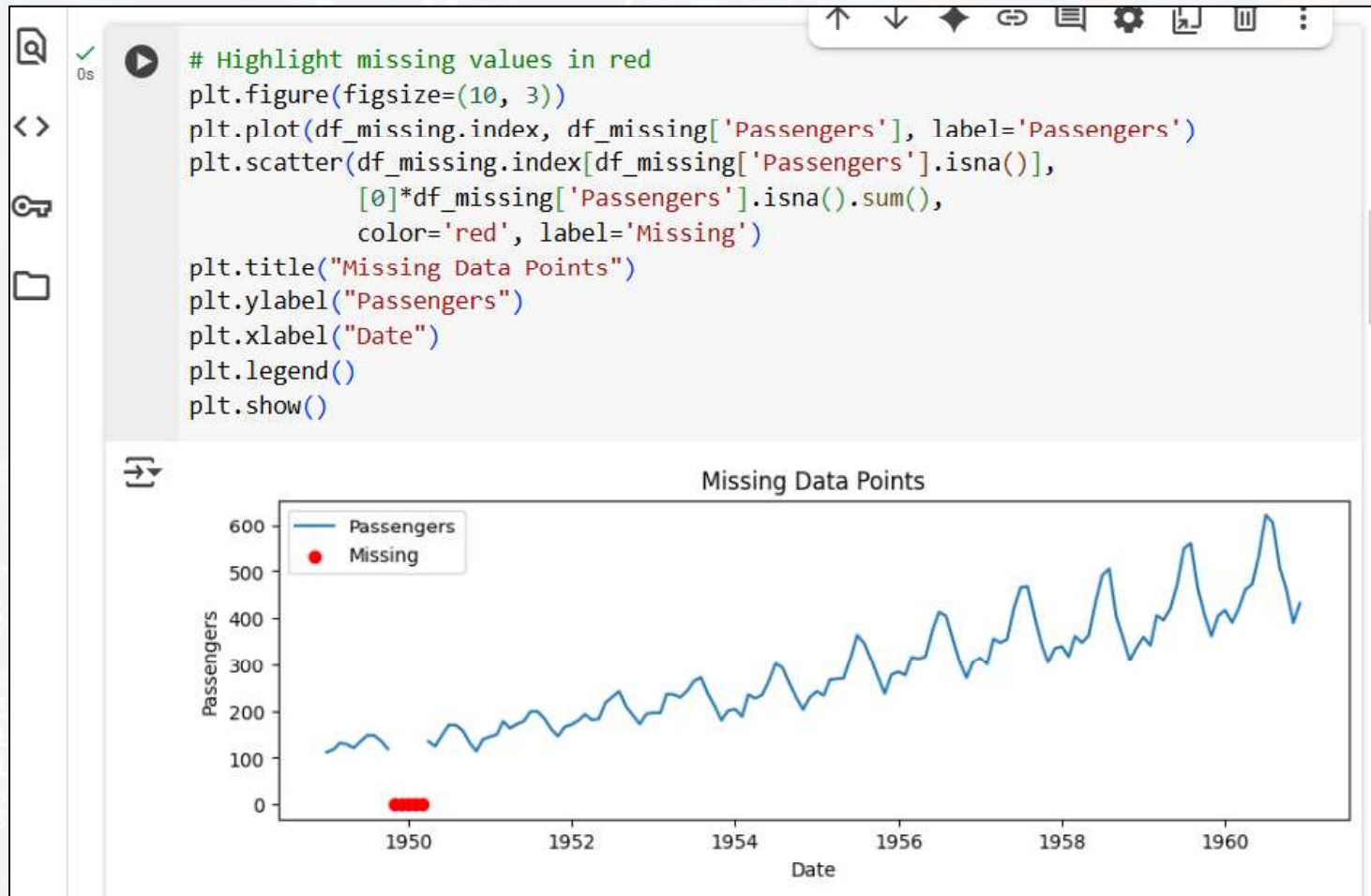
- **Plot 4: Residual**

- The random noise left after removing trend and seasonality
- Appears centered around 1 (because it's multiplicative), with no strong pattern, which is ideal

- **Data Visualisation to be covered in next section!!**



Data Cleaning: Missing Values





Data Cleaning: Missing Values

Handling Missing Data (If Any)

```
# Simulate missing values
df_missing = df.copy()
df_missing.iloc[5:8] = None

# Check missing values
print(df_missing.isna().sum())

# Fill missing values using forward fill
df_filled = df_missing.fillna(method='ffill')
```

```
Passengers    3
dtype: int64
```

Data Processing: Working with Dates

Parsing Dates and Setting Index

```
] # Convert 'Month' to datetime and set as index
df['Month'] = pd.to_datetime(df['Month'])
df.set_index('Month', inplace=True)

# Check info
df.info()
```

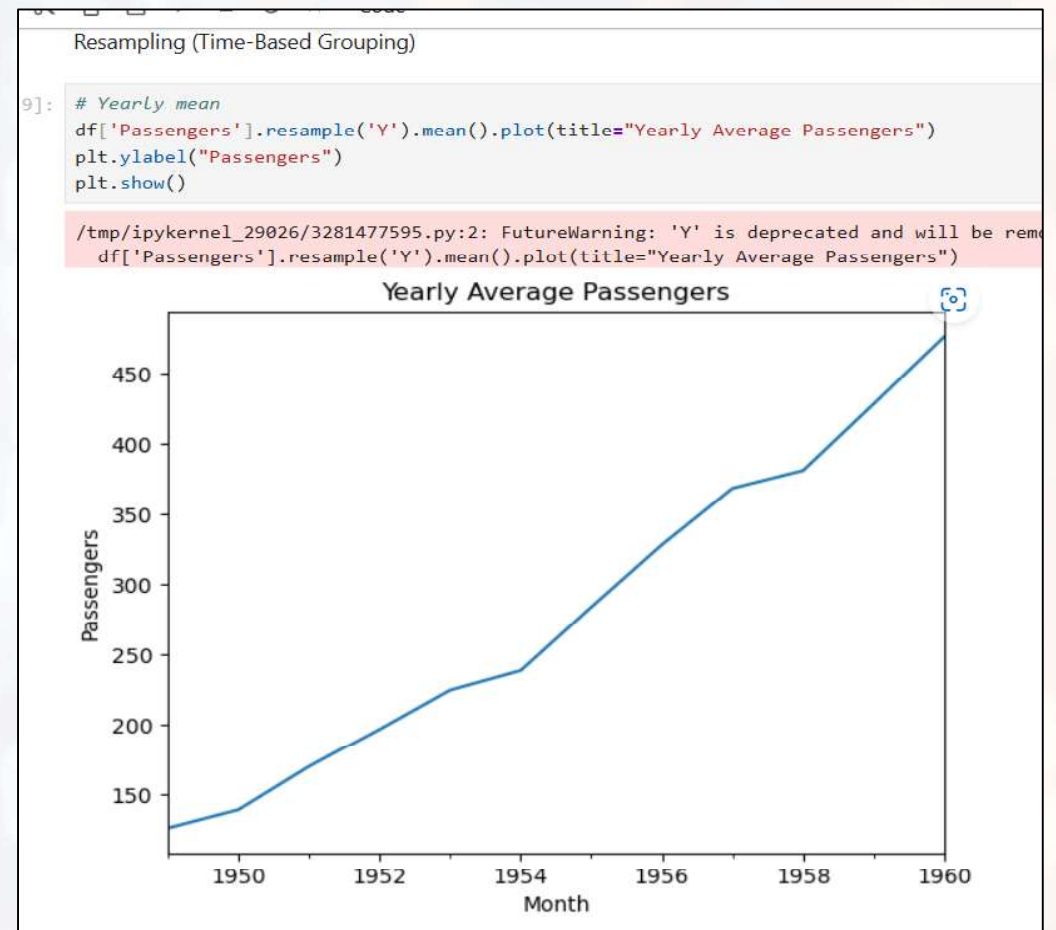
```
<class 'pandas.core.frame.DataFrame'>
DatetimeIndex: 144 entries, 1949-01-01 to 1960-12-01
Data columns (total 1 columns):
 #   Column      Non-Null Count  Dtype
---  ---
 0   Passengers  144 non-null    int64
dtypes: int64(1)
memory usage: 2.2 KB
```

Visualizing the Time Series

- **Parsing Dates:** Converts date strings (e.g., "1949-01") into proper datetime objects so Python can understand and work with them as actual dates.
- **Setting Index:** Makes the date column the index of the DataFrame, which is essential for time-based operations like resampling, rolling averages, and plotting trends over time.

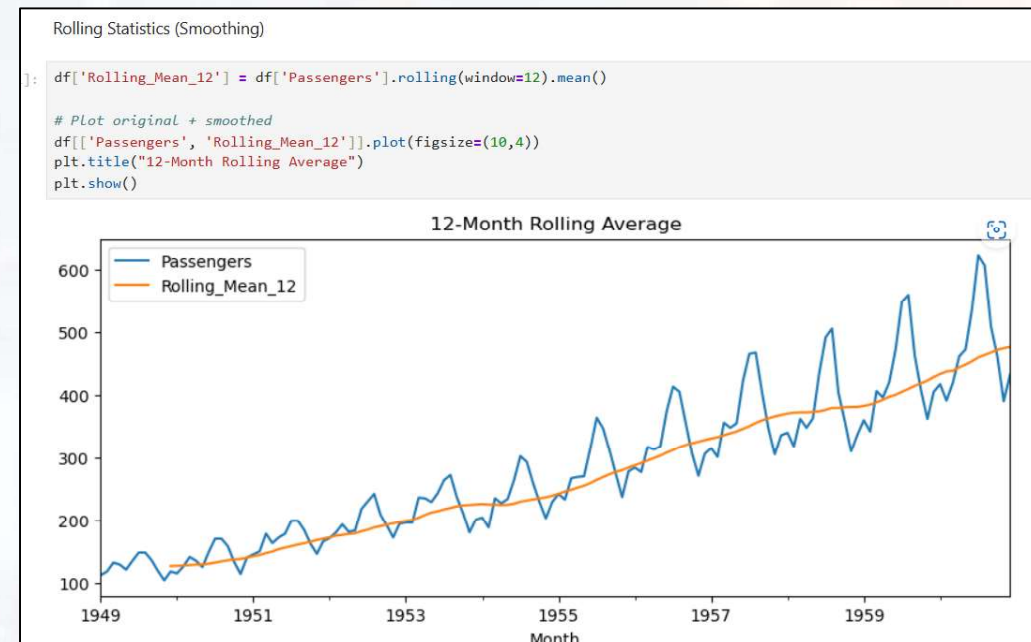
Data Processing

- Resampling (Time-Based Grouping)
- Resampling involves changing the frequency of your time series data — for example, converting monthly data to yearly data by averaging or summing values over each year.



Data Processing

- Rolling Statistics (Smoothing)
 - Rolling statistics calculate values (like mean or std) over a moving window of fixed size. This is used to smooth short-term fluctuations and highlight long-term trends.
 - window=12: Looks at 12 months (1 year)
 - Common for identifying trends in time series data
- This is a **processing technique** that supports **trend detection** and helps reduce visual noise in plots.



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Conclusion